or by transforming your quantitative data into categorical data using techniques like quantiles, which group numeric values.

For one of our datasets, we have readily accessible quantitative data: the population figures in the cities.csv table. For the tweets dataset, though, it seems like we don't have any quantitative data available, which is why we'll spend time in section 2.1.3 looking at how to transform data.

CATEGORICAL

Categorical data falls into discrete groups, typically represented by text, such as nationality or gender. Categorical data is often represented using shape or color. You map the categories to distinct colors or shapes to identify the pattern of the groups of elements positioned according to other attributes.

The tweets data has categorical data in the form of the user data, which you can recognize by intuitively thinking of coloring the tweets by the user who made them. Later, we'll discuss methods to derive categorical data.

TOPOLOGICAL

Topological data describes the relationship of one piece of data with another, which can also be another form of location data. The genealogical connection between two people or the distance of a shop from a train station each represent a way of defining relationships between objects. Topological attributes can be represented with text referring to unique ID values or with pointers to the other objects. Later in this chapter we'll create topological data in the form of nested hierarchies.

For the cities data, it seems like we don't have topological data. However, we could easily produce it by designating one city, such as San Francisco, to be our frame of reference. We could then create a distance-to-San-Francisco measure that would give us topological data if we needed it. The tweets data has its topological component in the `favorites` and `retweets` arrays, which provide the basis for a social network.

GEOMETRIC

Geometric data is most commonly associated with the boundaries and tracks of geographic data, such as countries, rivers, cities, and roads. Geometric data might also be the SVG code to draw a particular icon that you want to use, the text for a class of shape, or a numerical value indicating the size of the shape. Geometric data is, not surprisingly, most often represented using shape and size, but can also be transformed like other data, for example, into quantitative data by measuring area and perimeter.

The cities data has obvious geometric data in the form of traditional latitude and longitude coordinates that allow the points to be placed on a map. The tweets data, on the other hand, has no readily accessible geometric data.

TEMPORAL

Dates and time can be represented using numbers for days, years, or months, or with specific date-time encoding for more complex calculations. The most common